

Score: 95.0%
19/20 points

WISH - Digital - Day-3

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Q1. Which among the below is not a component of verification testbench?

- ☐ a. Driver
- ☐ b. Monitor
- ☐ c. Scoreboard
- ☐ **d. Test Pattern Generation**

1/1 point

Q2. When verifying an RTL for a simple calculator which performs basic operations (addition, subtraction, division, multiplication) and inputs only positive integers and the operator which among the below will not be a testcase?

- ☐ a. Tests for checking all operations
- ☐ b. Division by zero
- ☐ c. Test to check reset functionality

☐ **d. Operations on fractional numbers**

1/1 point

Q3. Which component of a verification testbench is responsible for checking the correctness of an observation?

☐ a. Driver

☐ c. Monitor

☒ **b. Scoreboard**

☐ d. Stimulus Generator

1/1 point

Q4. Which among the below is a component in SoC verification but not in IP verification?

☐ a. Formal Verification

☒ **c. AMS Co-sims**

☐ b. GLS

☐ d. None

1/1 point

Q5. What is the full-form for GLS?

☒ **a. Gate Level Simulation**

☐ c. Gate Level System

☐ b. Gate Layer Simulation

☐ d. Gate Layer System

1/1 point

Q6. Which component of verification helps evaluate the progress of verification

- ☐ a. Testplan Creation ☐ c. Checkers
- ☐ **b. Coverage** ☐ d. GLS

1/1 point

Q7. Which category of coverage has the requirements defined by the engineer on the basis of a Product Specification?

- ☐ a. Code Coverage ☐ **c. Functional Coverage**
- ☐ b. FSM Coverage ☐ d. Statement Coverage

0/1 point

Q8. There is a bug in the RTL code of a 4-bit adder (inputs being a and b) where one of the inputs (input a) is defined to be of 3 bits instead of 4. Which case among the below will catch the bug?

- ☐ **a. Range Check**
- ☐ **b. Overflow Check**
- ☐ c. Addition with a=7 and b=9
- ☐ d. Addition of numbers constrained to 7

2/2 points

Q9. A calculator has a bug where an input of zero is internally treated as 1 and input of 1 is internally treated as a 0. With this information, what will be the result of the following operation: 91×10

☐ a. 910

☐ c. 991

☐ b. 990

☒ d. **90**

1/1 point

Q10. A calculator has a bug where an input of zero is internally treated as 1 and input of 1 is internally treated as a 0. With this information, what will be the result of the following operation: $101010/3$

☐ a. 37037

☒ c. **3367**

☐ b. 33670

☐ d. 0

1/1 point

Q11. There is a bug in a 4-bit counter where the index 1 of the output is tied off to zero. Which of the following operations will give the wrong sum value:

☒ a. **'b1111 + 'b1000**

☐ c. 'b1010 + 'b1010

☐ b. 'b1001 + 'b1000

☐ d. 'b0100 + 'b1000

1/1 point

Q12. A calculator has a bug which makes it output 4 instead of a 3 and 3 instead of 4. What will be seen on screen as an output when the following operation is carried out: 31×33

☐ a. 1023

☒ b. 1024

☐ c. 1804

☐ d. 1364

1/1 point

13. A calculator has a bug which makes it output 4 instead of a 3 and 3 instead of 4. What will be seen on screen as an output when the following operation is carried out: $21717 + 21716$

☐ a. 43433

☐ c. 43444

☒ b. 34344

☐ d. 34343

1/1 point

14. If your design fails only for random inputs but passes directed tests, what is likely?

☐ a. Design is perfect

☐ c. Simulator issue

☒ b. Missing corner cases

☐ d. Clock Problem

1/1 point

15. Can 100% coverage guarantee a bug-free design?

☐ a. Yes

☒ **b. No**

1/1 point

16. What is the role of a monitor in a verification testbench?

☐ a. Drives signals

☐ c. Generates testcases

☒ **b. Observes DUT activity**

☐ d. Synthesizes logic

1/1 point

17. If a test fails, what is the FIRST step?

☐ a. Rewrite the design

☒ **c. Analyze the logs and waveforms**

☐ b. Ignore the failure

☐ d. Change constraints randomly

1/1 point

Q19. What happens if signal widths mismatch unintentionally?

☐ a. No issue

☐ c. Faster simulation

☒ **b. Possible truncation or extension bugs**

☐ d. Better coverage

1/1 point

Q20. Race condition occurs when:

- ☐ a. Signals toggle fast
- ☐ b. Clock is unstable
- ☒ c. **Multiple events compete in the same timestep**
- ☐ d. Coverage is low

1/1 point

Done



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